higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE

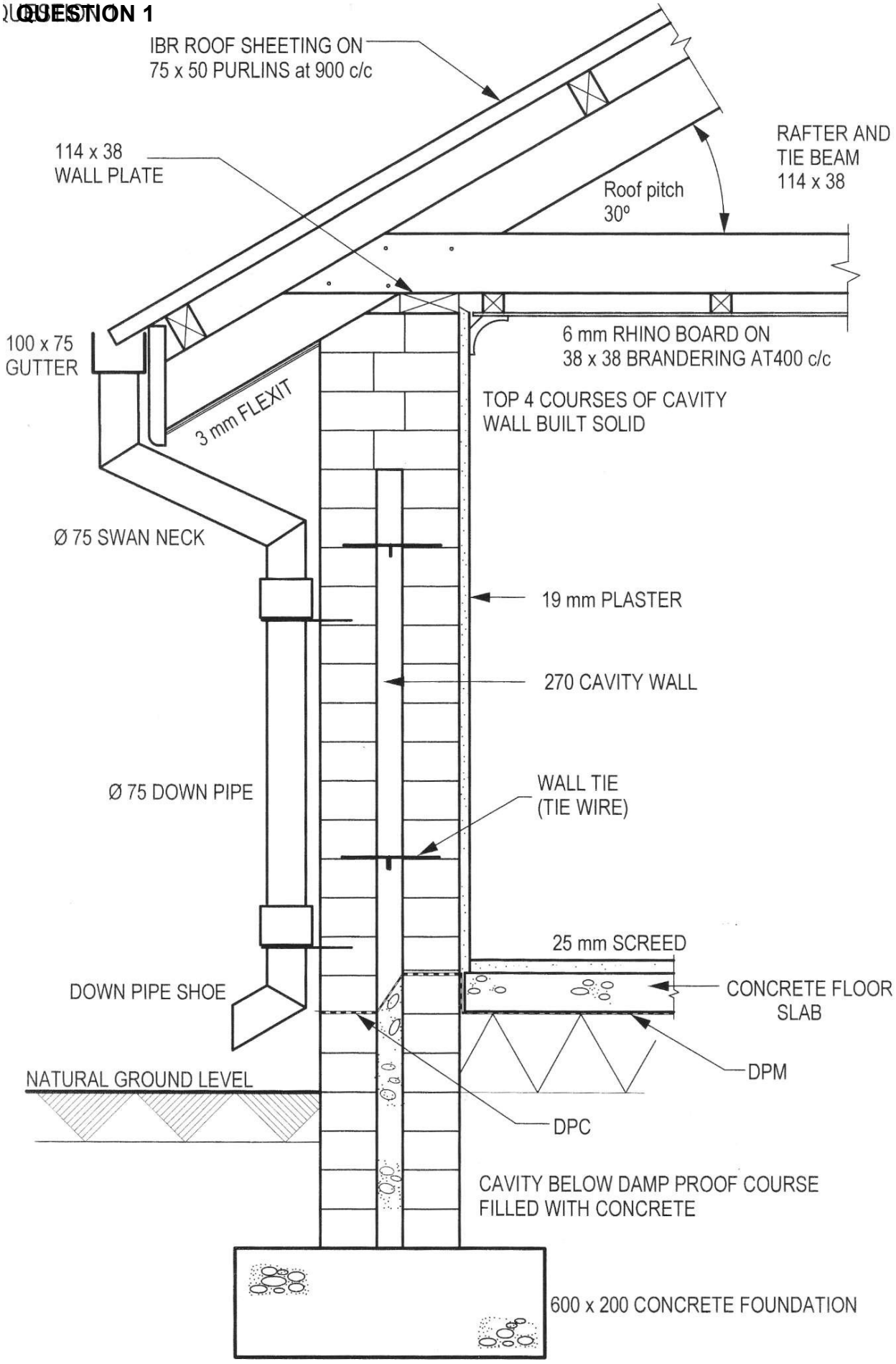
BUILDING AND STRUCTURAL CONSTRUCTION N4

9 APRIL 2019

This marking guideline consists of 9 pages.

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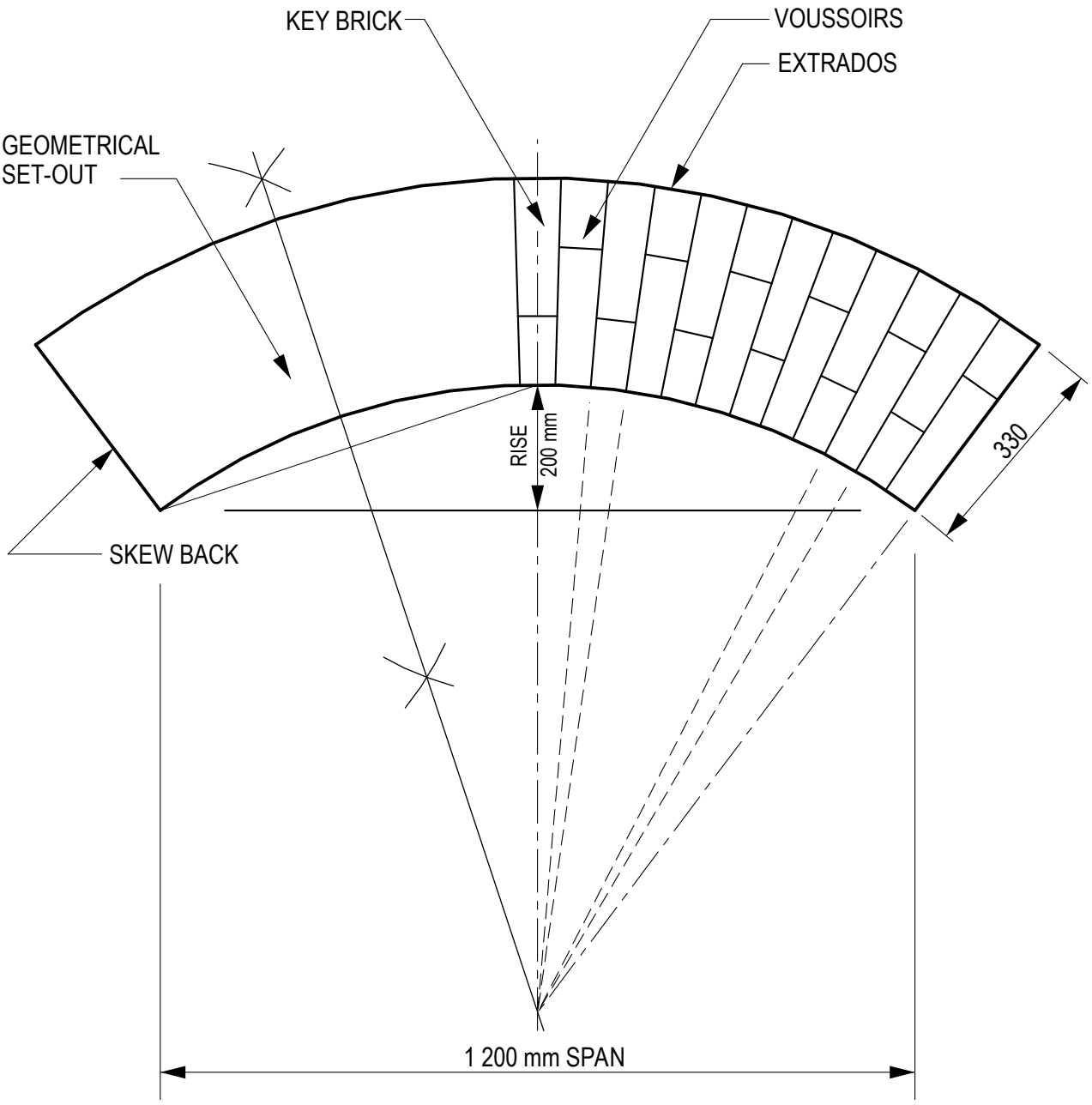
QUESTION 1



[22]

VERTICAL SECTION THROUGH THE EAVES OF A HOUSE
SCALE 1:10

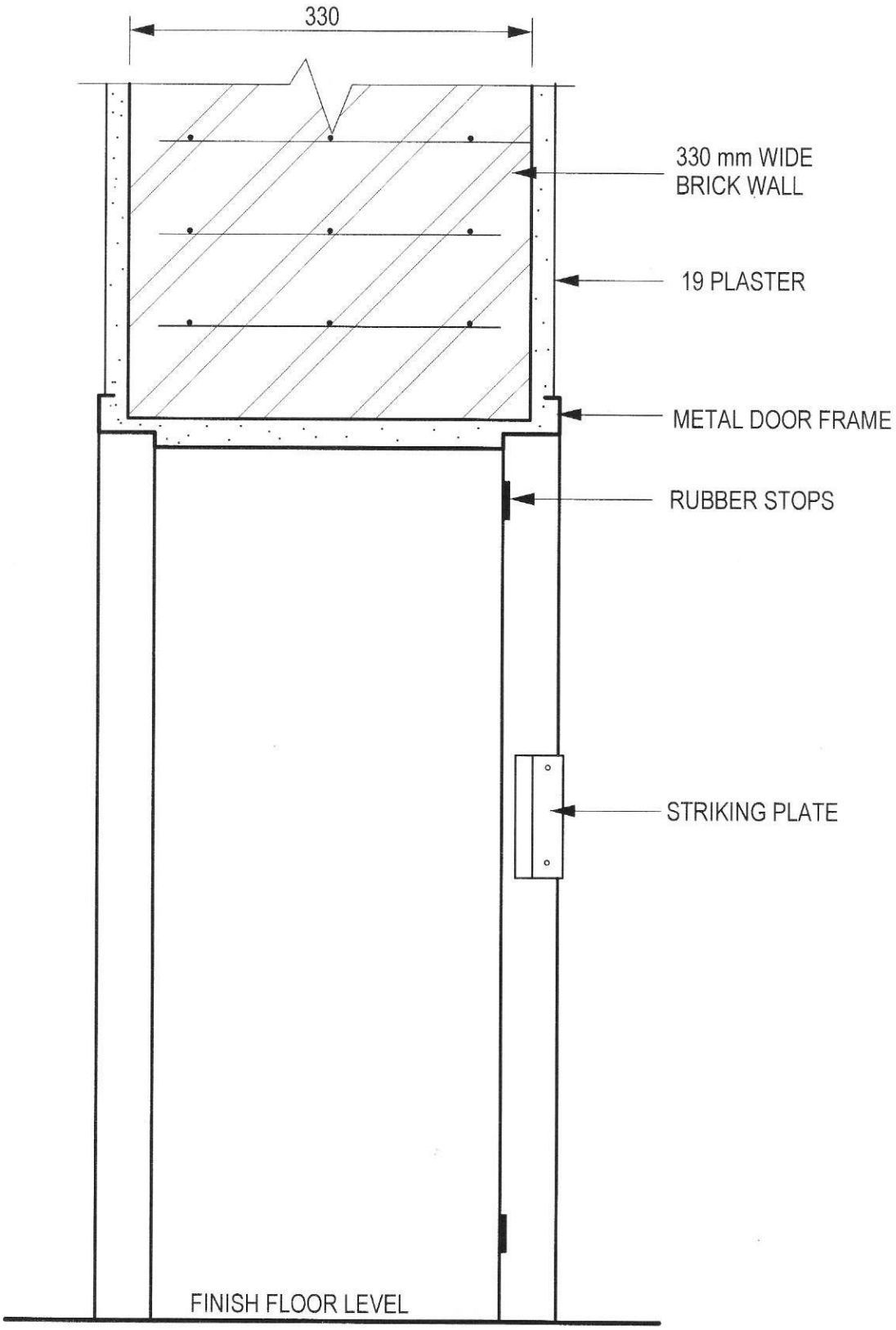
QUESTION 2



SEGMENTAL GAUGED ARCH
SCALE 1:10

[12]

QUESTION 3



VERTICAL SECTION THROUGH STEEL DOORFRAME
SCALE 1:5

[11]

QUESTION 4

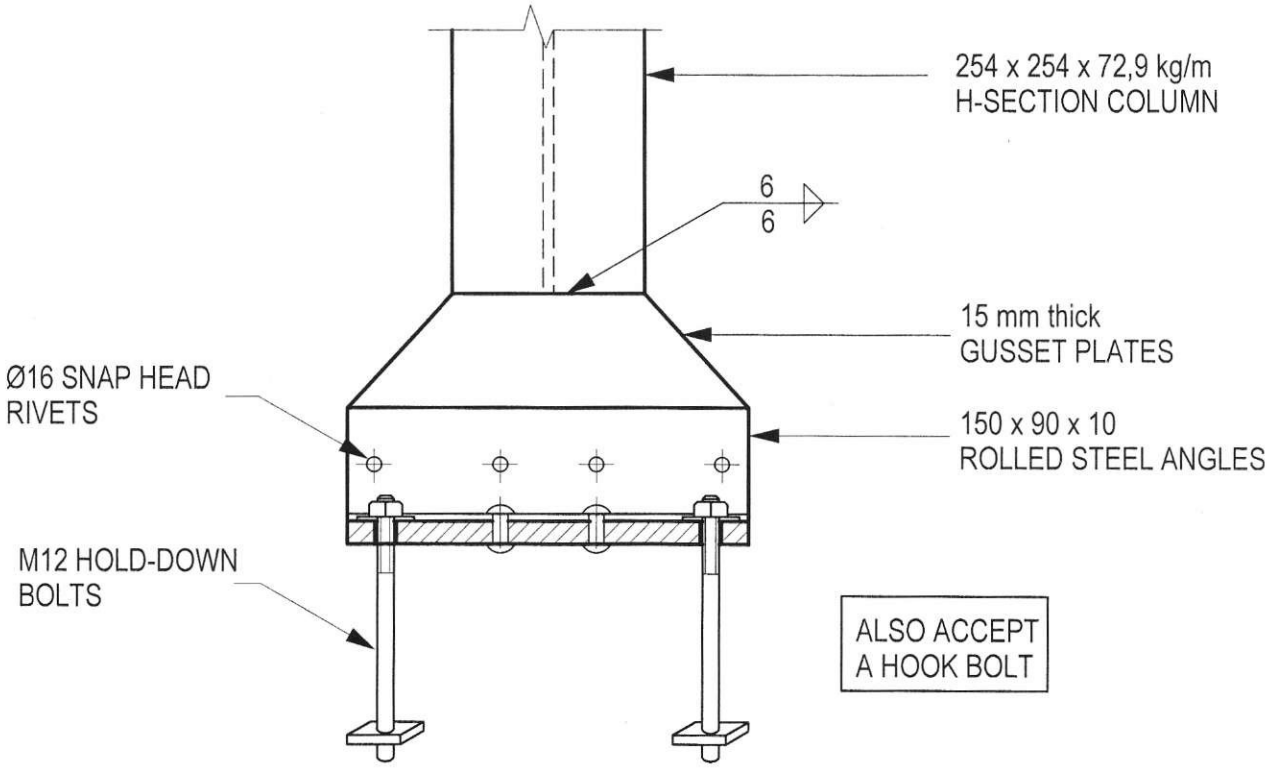
4.1 4.1.1 Ridge capping/Ridging
4.1.2 Apron/Counter flashing
4.1.3 Head-wall flashing

(3 × 1) (3)

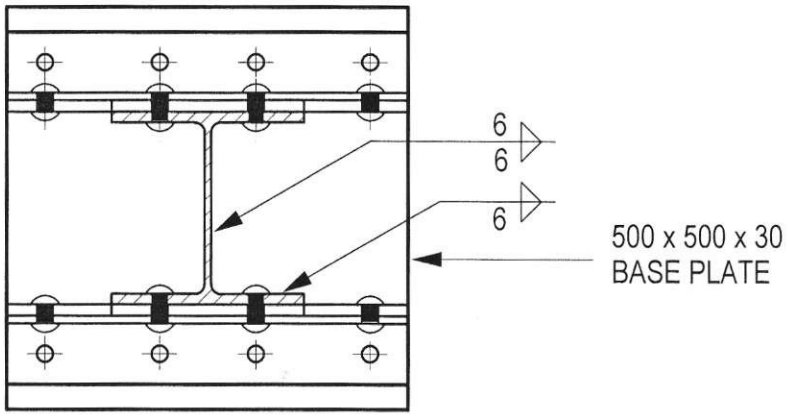
4.2 Centres are temporary wooden structures constructed to support brick, concrete or stone arches during construction. (2)

4.3 A suspended ceiling is fixed to a framework suspended from the main structure to form a void between the two components. The ceiling can also be used to decrease the volume of a room. (2)
[7]

QUESTION 5



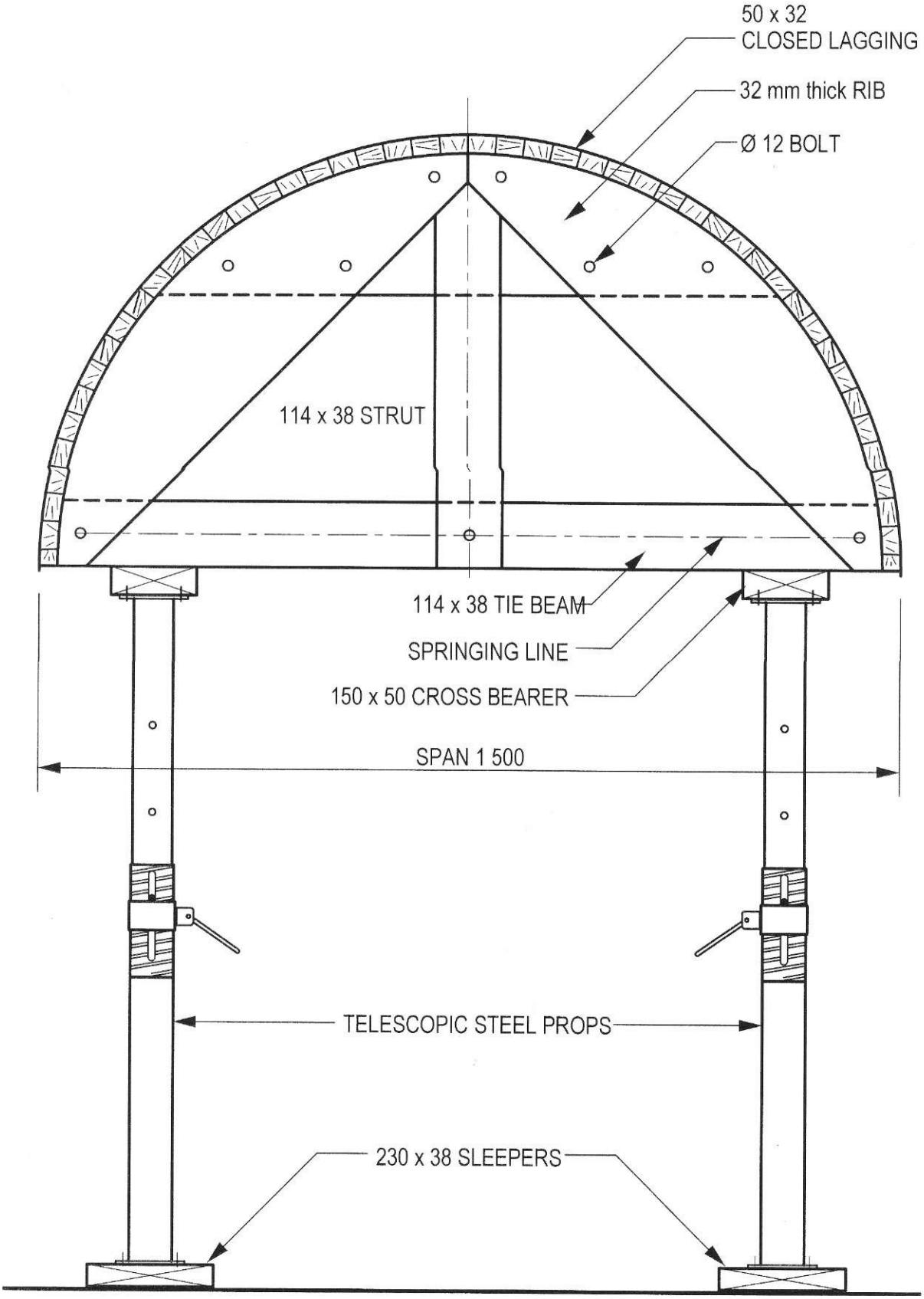
VERTICAL SECTION THROUGH THE STEEL COLUMN AND BASE PLATE
SCALE 1:10



TOP VIEW OF A STEEL COLUMN AND BASE PLATE
SCALE 1:10

[18]

QUESTION 6

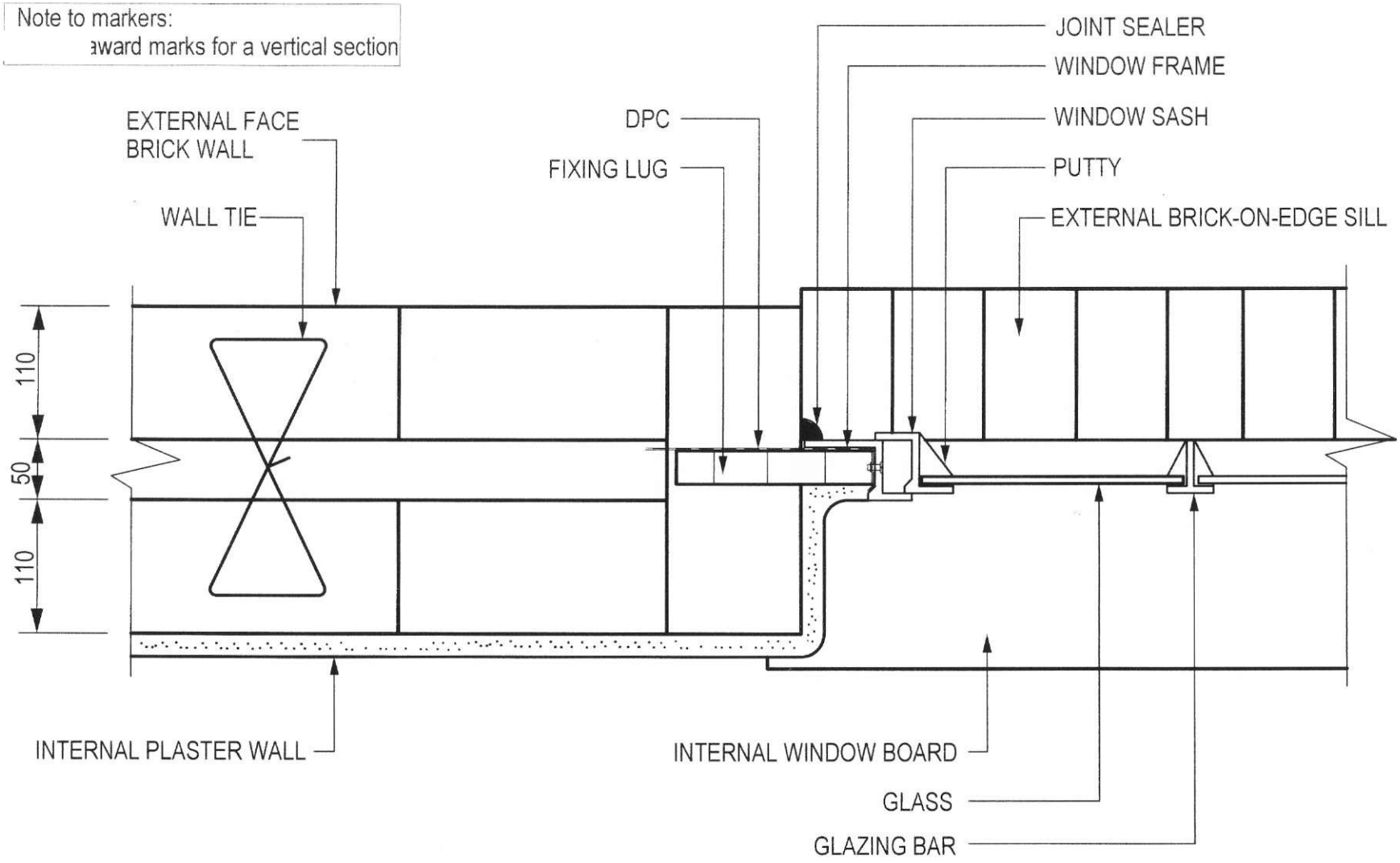


CENTRING FOR A LIGHT SEMICIRCULAR ARCH
SCALE 1:10

[15]

QUESTION 7

Note to markers:
award marks for a vertical section



HORIZONTAL SECTION THROUGH A STEEL WINDOW
SCALE 1:2

[15]

MARK ALLOCATION

ANY ALTERNATIVE CORRECT ANSWER MUST BE CONSIDERED CORRECT

QUESTION 1		
MARK ALLOCATION		
CORRECTNESS:		15
Conc Fndt	1	
Cavity wall	2	
Floor Construc.	2	
Roof Construc.	3	
Ceiling Construc.	2	
Gutter& Down pipe.	2	
Plaster & Screed	2	
Ground Level	1	
Accuracy		3
Hatching		1
Labels		3
Total		22

QUESTION 2		
MARK ALLOCATION		
Correctness:		7
Span & Rise corr	1	
Geom set-out	2	
Skew backs corr	1	
330 mm deep corr	1	
Bricks corr.	2	
Accuracy		1
Labels		2
dimensions		2
Total		12

QUESTION 3		
MARK ALLOCATION		
Correctness:		7
Wall thickness corr.	1	
Sectional view of frame.	2	
Stops & Plate	2	
Brickforce	1	
Plaster	1	
Accuracy		2
Dimension		1
Labels		1
Total		11

QUESTION 5			
MARK ALLOCATION		MARK ALLOCATION	
Correctness: (Q5.1)		Correctness: (Q5.2)	
Column in place	1	Base Plate	1
Gusset plate	1	Column	1
Angle	1	Gussets	1
Rivets	1	Angles	1
Hook Bolts	2	Rivets	1
Total	6	Total	5

MARK ALLOCATION	
Accuracy	3
Weld Symbols	2
Lables	2
Total	7

SUMMARY
6 + 5 + 7
Total = 18 Marks

QUESTION 6		
MARK ALLOCATION		
Correctness:		10
Span corr	1	
Centering	3	
Lagging	1	
Steel Props	3	
Bearers	1	
Sleepers	1	
Accuracy		3
Labels		2
Total		15

QUESTION 7		
MARK ALLOCATION		
Correctness:		9
Brick wall width corr.	1	
Window sections corr.	2	
Glazing bar	1	
Glass & putty	1	
Window sills	1	
Fixing lug	1	
DPC	1	
Tie wire	1	
Accuracy		2
Plaster		1
Labels		3
Total		15

TOTAL: 100

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